

Sagar Sinha

(602) 670-3742 | sinhasagar507@gmail.com | linkedin.com/in/sagarsinha2000 | github.com/sinhasagar507

EDUCATION

MS Computer Science

Arizona State University

Dec 2025

GPA: 3.93/4.0

- **Relevant Coursework:** Data Mining, Applied Regression Analysis, NLP, Data Visualization, Distributed Database Systems

BS Computer Engineering

Pandit Deendayal Energy University

May 2022

GPA: 9.02/10.0

- **Relevant Coursework:** Data Structures, Algorithms, RDBMS, Data Warehousing, Statistics, Machine Learning

PROFESSIONAL EXPERIENCE

AWS Data Developer Intern, MindSparked

Mar 2026 – Present

- Built AWS-based data workflows for Kairos, an AI-enabled education platform, using Lambda, S3, Step Functions, API Gateway, and Aurora PostgreSQL to support reporting, monitoring, and internal analytics operations.
- Reduced data retrieval latency by 40% through optimization of Aurora PostgreSQL query paths, removal of redundant SQL joins, and addition of targeted indexes for high-frequency analytics workflows.
- Improved backend data observability by analyzing CloudWatch logs, Lambda invocation patterns, RDS-processing activity, and Aurora workload metrics to identify bottlenecks and reliability risks.
- **Technical Skills:** SQL, PostgreSQL, Python, AWS Lambda, S3, Aurora RDS, Step Functions, API Gateway, CloudWatch, Terraform

AI Intern, Extern

Sep 2025 – Dec 2025

- Engineered the core document retrieval engine for an AI Mortgage Assistant, transforming unstructured data into searchable context to improve User Discovery Success and reduce time-to-information for loan officers.
- Optimized search relevance (MRR of 0.98) and reduced user friction by achieving a 2s median query latency, directly improving the Product Task Completion Rate for complex document inquiries.
- Developed product health monitoring by implementing automated data observability and logging, enabling real-time tracking of document processing success and feature adoption metrics.
- **Technical Skills:** Python (PyMuPDF, PaddleOCR), Vector Databases, Ranking, Natural Language Processing, Information Retrieval, Large Language Models (LLMs), Mortgage Industry

Data Analyst, Arizona State University

Jul 2024 – Dec 2025

- Extracted and standardized 50K+ records by designing SQL ETL pipelines to significantly enhance data quality and accuracy for downstream reporting.
- Visualized campus health KPIs for leadership stakeholders through interactive Tableau and Excel dashboards, enabling data-driven root cause analysis of health trends.
- Reduced manual data manipulation time by 4x while ensuring reliable reporting inputs by establishing automated QA checks and organizational data governance protocols.
- **Technical Skills:** SQL, Tableau, ETL, Data Governance, KPI Reporting, Excel

Data Science Researcher, Accredian

Apr 2023 – Jul 2023

- Optimized engagement strategies for 3000+ learners by conducting Python-based Exploratory Data Analysis (EDA) and feature engineering on behavioral datasets.
- Resolved data discrepancies for non-technical stakeholders by serving as the Subject Matter Expert for database troubleshooting and technical communications.
- **Technical Skills:** Python (Pandas, Matplotlib), Statistical Analysis, Stakeholder Communication, Project Management

RELEVANT PROJECTS

Enterprise Sales Data Warehouse

- **Streamlined visibility into sales, products, and customer segmentation** by architecting a Snowflake warehouse using Medallion architecture to unify ERP and CRM data
- **Ensured data integrity for executive-level revenue reporting** by automating SQL ETL pipelines and governance protocols to standardize complex data transformation.
- **Technical Skills:** Database Architecture Design, Metric Development, ETL architecture, Snowflake SQL

Sales Pulse

- **Provided real-time, interactive insights for stakeholders** by administering data cleaning, transformation, and modeling in Tableau using advanced calculations and parameters.
- **Visualized \$609K in total sales and a 32.7% year-over-year profit increase** through dynamic dashboards that reduced manual reporting time by 25% and enabled real-time monitoring of monthly performance trends.
- **Technical Skills:** Tableau (Prep Builder, KPI Development, Charts, Dashboards)

Urban Mobility Insights

Spring 2025

- Utilized **dbt** to engineer a **STAR schema** in **Bigquery** for integrating taxi and climate datasets, reducing **ELT latency by 30%** and accelerating KPI exploration for urban planning.
- Orchestrated a **Spark** pipeline to process **10M+** NYC taxi records, engineering spatiotemporal features to predict trip durations and optimize service-level mobility insights.
- **Technical Skills:** Python, Apache Airflow, Google Cloud (GCS, Bigquery, Dataproc),